Lesson notes · for you

# Repeat it: loops

Programming · Level 1 · Lesson 2

**Time**
20 min

**Materials**
Paper and a pencil; Arrow cards and number cards 1 to 4; A phone (optional)

## Learning goals

- Spot steps that repeat.
- Use a loop to make a program shorter.
- Count cards to find the shortest program.

## Activity: Activity: Clap the loop

1. Warm-up (3 min): Clap “clap, clap, stamp” four times. How would you tell someone to do it, in as few words as possible?
2. Write (7 min): Write the pattern in full (12 moves), then as a loop: “clap, clap, stamp: 4 times”. Count the words in each.
3. Create (7 min): Make up your own pattern of three moves and write it with a loop.
4. Share (3 min): Give your loop to a partner. Can they perform it exactly?

**No device is needed: hands, feet and paper are enough.**

## Discussion

- Why is the loop easier to change?
- Which songs or games have loops?
- When would you not use a loop?

## Check yourself

- Write one thing you do again and again as a loop.

**Try it this week**
Teach someone at home a clapping pattern using only a loop: “clap, clap, stamp: 3 times”. Did they do it right?

Worksheet

# Repeat it: loops

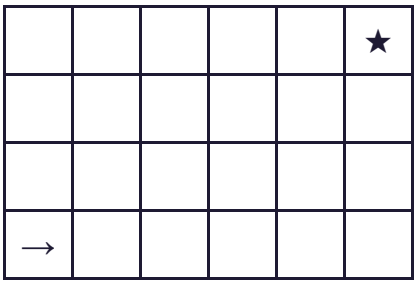
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Name _____ Class _____ Date _____

**1** **Which program is shorter, but walks the same way?**

◯ Forward, forward, forward ◯ Forward, three times ◯ They are the same length

**2** **Walk the robot to the flag with 5 cards at most. Without loops you would need 9!**




Cut out cards (↑ forward, ↶ turn left, ↷ turn right) and lay out a program that takes the robot to the ★. Write it here:

**3** **Match each long program to its loop.**

| | | |
|---|---|---|
| forward, forward, forward, forward | • • • • | jump × 3 |
| turn right, turn right | • • • • | turn right × 2 |
| jump, jump, jump | • • • • | forward × 4 |

**4** **A program: forward × 3, turn right, forward × 2. How many cards does it use?**

◯ 3 ◯ 5 ◯ 6

**5** **"Turn right × 4" leaves the robot facing the same way as when it started.**

◯ True ◯ False

✎ **Find a loop in your day: something you do again and again. How many times do you repeat it?**